Chapter 12. Identifiable Causal Discovery Models: LiNGAM

Graphical models, Lecture 5 (non-Gaussian errors, residual independence, ICA-LiNGAM, DirectLiNGAM, variants)

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1 Directions That Conditional Independence Cannot Distinguish

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

All three satisfy $X \perp\!\!\!\perp Z \mid Y$ (Ch. 9). In

$$X = BX + \epsilon, \qquad \epsilon_j \text{ independent},$$

the error *distribution*, not only its covariance, can carry direction information.

2 The LiNGAM Model

Motivation. With non-Gaussian noise only the true direction leaves a residual independent of the regressor, which identifies the causal order.

Definition: LiNGAM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad X = BX + \epsilon.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering of variables, B is strictly lower triangular
independent errors	$\epsilon_1, \ldots, \epsilon_p$ are mutually independent: no hidden common cause among the analyzed variables
non-Gaussianity	the structural errors are not normally distributed, which leaves information that distinguishes directions

Theorem: identifiability of LiNGAM (Shimizu et al., 2006)

Under linearity, acyclicity, mutually independent errors and non-Gaussian errors (at most one Gaussian error is allowed), the coefficient matrix B — hence the DAG and the causal order — is uniquely determined by the distribution of X.

3 In the True Direction, Cause and Regression Residual Are Independent

3.1 Two variables

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X \perp\!\!\!\perp \epsilon_Y, \quad \epsilon_X, \epsilon_Y \text{ non-Gaussian}.$$

$r_Y = Y - bX = \epsilon_Y$, so $X \perp\!\!\!\perp r_Y$.

Reverse $Y \to X$: $X = cY + r_X$; Y and r_X both contain ϵ_X, ϵ_Y, so typically $Y \not\perp\!\!\!\perp r_X$.

Theorem: Darmois–Skitovich

Let $\epsilon_1, \dots, \epsilon_m$ be independent and $U = \sum_i \alpha_i \epsilon_i$, $V = \sum_i \beta_i \epsilon_i$. If $U \perp\!\!\!\perp V$, then every ϵ_i with $\alpha_i \beta_i \neq 0$ is Gaussian.

Theorem: the reverse regression leaves a dependent residual

In the model above with $b \neq 0$ and $\sigma_X^2 = \mathrm{Var}\epsilon_X$, $\sigma_Y^2 = \mathrm{Var}\epsilon_Y$, the least-squares coefficient is $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = b\sigma_X^2/(b^2\sigma_X^2 + \sigma_Y^2)$ and

$$r_X = X - cY = \frac{\sigma_Y^2}{b^2\sigma_X^2 + \sigma_Y^2}\,\epsilon_X - c\,\epsilon_Y, \qquad Y = b\,\epsilon_X + \epsilon_Y.$$

r_X is uncorrelated with Y but, if either error is non-Gaussian, not independent of it.

Proof [Supplement]. $r_X = (1-cb)\epsilon_X - c\epsilon_Y$ with $1 - cb = \sigma_Y^2/(b^2\sigma_X^2 + \sigma_Y^2)$; all coefficients of r_X and Y are nonzero, so $r_X \perp\!\!\!\perp Y$ would force both errors Gaussian (Darmois–Skitovich). Uncorrelatedness is the normal equation. ■

With Gaussian errors, (r_X, Y) uncorrelated $\Rightarrow$ independent in *both* directions: no identifiability. **LiNGAM picks the direction with the more independent residual.**

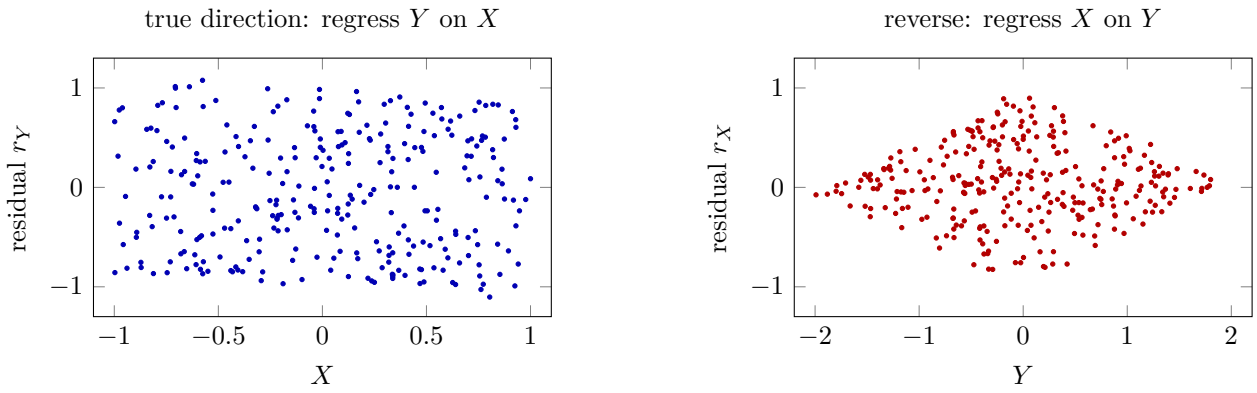


Figure 1: $X = \epsilon_X$, $Y = X + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \mathrm{Unif}(-1, 1)$, $n = 300$. True direction: rectangle (independent); reverse: diamond, uncorrelated but dependent (correlation of squares 0.07 vs −0.42).

Example: a simulation picture

With $\mathrm{Unif}(-1, 1)$ errors, (X, r_Y) fills a rectangle; (Y, r_X) fills a diamond whose spread of r_X shrinks at the extremes of Y.

4 Estimation

4.1 ICA-LiNGAM

$X = (I - B)^{-1}\epsilon = A\epsilon$ is a mixture of independent non-Gaussian sources (ICA).

Algorithm: ICA-LiNGAM

1. Run ICA to estimate an unmixing matrix W with WX having independent components; W equals $I - B$ up to an unknown row permutation and row scaling.

2. Find the row permutation of W giving no zeros (in practice, no small entries) on the diagonal; it is unique (Exercise 12.1).
3. Divide each row by its diagonal element to get W', and set $\hat{B} = I - W'$.
4. Find the causal order by permuting $\hat{B}$ to be as close as possible to strictly lower triangular; prune small coefficients (e.g. by bootstrap or adaptive Lasso).

ICA is non-convex: local optima, dependence on initial values.

4.2 DirectLiNGAM finds exogenous variables one by one

Theorem: characterization of an exogenous variable

In a LiNGAM, let $r_i^{(j)} = X_i - \frac{\mathrm{Cov}(X_i, X_j)}{\mathrm{Var}(X_j)} X_j$ for $i \neq j$. Then X_j is exogenous (has no parents) iff $X_j \perp\!\!\!\perp r_i^{(j)}$ for all $i \neq j$.

Proof ($\Rightarrow$) and idea ($\Leftarrow$) [Supplement]. ($\Rightarrow$) $X_j = \epsilon_j$, $X_i = a_{ij}\epsilon_j + \sum_{l \neq j} a_{il}\epsilon_l$ with $A = (I-B)^{-1}$, and $\mathrm{Cov}(X_i, X_j)/\mathrm{Var}(X_j) = a_{ij}$, so $r_i^{(j)} = \sum_{l \neq j} a_{il}\epsilon_l \perp\!\!\!\perp \epsilon_j$. ($\Leftarrow$) If X_j has a parent, some $r_i^{(j)}$ shares a non-Gaussian error with X_j with nonzero coefficients; Darmois–Skitovich gives dependence (Shimizu et al., 2011). ■

The residuals $(r_i^{(j)})_{i \neq j}$ again follow a LiNGAM (Exercise 12.3), so the step repeats.

Algorithm: DirectLiNGAM

For each candidate X_j, regress every other X_i on X_j to obtain residuals $r_i^{(j)}$, and add up the dependence between the candidate and all residuals:

$$T(j) = \sum_{i \neq j} \mathrm{Dep}(X_j, r_i^{(j)}).$$

1. Evaluate every variable as an exogenous candidate.
2. Choose the variable with the smallest $T(j)$ as the first in the order.
3. Replace the remaining variables by the residuals with the chosen variable's effect removed.
4. Repeat the same process on the remaining variables.

Dep: kernel independence measure or mutual information. No non-convex step; with a consistent Dep it finds the order in $p - 1$ steps.

4.3 Repeating residual independence yields the causal order

Example: three variables

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3,$$

with independent non-Gaussian errors.

Step	Independence observed	Choice
1	X_1 is independent of the residuals of X_2, X_3 after removing X_1	choose X_1 first
2	after removing X_1's effect, X_2 is independent of the residual of X_3	choose X_2
3	the last remaining variable	choose X_3

Result: $X_1 < X_2 < X_3$ (in step 1, $T(2) > T(1)$ since the residual of X_1 on X_2 shares ϵ_1, ϵ_2 with X_2).

4.4 After the order: estimate direct edges and effects

With order $X_1 < X_2 < X_3$, earlier variables are candidate parents:

$$X_2 = \beta_{21} X_1 + \epsilon_2, \qquad X_3 = \beta_{31} X_1 + \beta_{32} X_2 + \epsilon_3.$$

$\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{32} \approx 0.7$, $\hat{\beta}_{31} \approx 0$:

$$X_1 \Rightarrow X_2 \Rightarrow X_3.$$

Least squares within the order, pruning by adaptive Lasso or tests; acyclicity is guaranteed.

5 Variants for Different Data Structures

Data or problem	Method	Caveat
independent cross-sectional data	DirectLiNGAM, ICA-LiNGAM	linear, acyclic structure, no hidden confounding
repeated measurements over time	VAR-LiNGAM	distinguish lagged and contemporaneous relations
feedback cycles	cyclic LiNGAM family	the acyclicity assumption of basic LiNGAM cannot be used
hidden common causes possible	ParceLiNGAM, RCD	estimate the identifiable part of the relations rather than the whole DAG

Remark: when LiNGAM is fragile

Near-Gaussian errors need large samples; measurement error, nonlinearity, confounding or selection can favour the wrong direction. Bootstrap the order and test final residual independence.

Summary

- Non-Gaussian independent errors break ties within the Markov equivalence class.
- True direction: cause $\perp\!\!\!\perp$ residual; reverse: uncorrelated but dependent (Darmois–Skitovich); Gaussian errors: no difference.
- ICA-LiNGAM reads B off the unmixing matrix; DirectLiNGAM peels off exogenous variables, then estimates coefficients.

6 Selected exercises

Motivation. The unique permutation in ICA-LiNGAM, the exact dependence in the figure, and the DirectLiNGAM recursion.

Exercise 12.1 (The row permutation in ICA-LiNGAM is unique)

Let $M = I - B$, where the variables admit a causal order κ (a bijection onto $\{1, \dots, p\}$) with $B_{jk} \neq 0 \Rightarrow \kappa(k) < \kappa(j)$.
(a) Show that the identity is the only permutation σ with $M_{\sigma(i),i} \neq 0$ for all i.
(b) Let $W = \Lambda P M$ with Λ diagonal invertible and P a permutation matrix (the ICA indeterminacy). Show that exactly one row permutation of W has no zero on the diagonal, and that it recovers M after dividing each row by its diagonal entry.

Solution [Supplement].

(a)

Step 1. $M_{jk} \neq 0 \Rightarrow j = k$ or $B_{jk} \neq 0$, hence $\kappa(k) \leq \kappa(j)$.

Step 2. So $\kappa(i) \leq \kappa(\sigma(i))$ for all i, and $\tau = \kappa \circ \sigma \circ \kappa^{-1}$ has $\tau(m) \geq m$; as $\sum_m (\tau(m) - m) = 0$, τ, hence σ, is the identity.

(b)

Step 3. Row i of W is λ_i times row $\pi(i)$ of M. Reordering by q puts $\lambda_{q(i)} M_{\pi(q(i)),\cdot}$ in row i, with diagonal entry $\lambda_{q(i)} M_{\pi(q(i)),i}$.

Step 4. All nonzero iff $\sigma = \pi \circ q$ has $M_{\sigma(i),i} \neq 0$ for all i, i.e. $q = \pi^{-1}$ by (a). This gives $\Lambda' M$; $M_{ii} = 1$, so row normalization returns M and $\hat{B} = I - M = B$. ■

Exercise 12.2 (The dependence in the simulation figure)

Let $X = \epsilon_X$ and $Y = X + \epsilon_Y$ with $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$.
(a) Show that in the true direction $\text{Cov}(X^2, r_Y^2) = 0$.
(b) For the reverse regression $r_X = X - cY$, show that $c = \frac{1}{2}$, $\text{Cov}(r_X, Y) = 0$ and $\text{Corr}(r_X^2, Y^2) = -\frac{3}{7}$.

Solution [Supplement]. Let $u = \epsilon_X$, $v = \epsilon_Y$: $\mathbb{E}u = \mathbb{E}u^3 = 0$, $\mathbb{E}u^2 = \frac{1}{3}$, $\mathbb{E}u^4 = \frac{1}{5}$, likewise v.

(a) $r_Y = v \perp\!\!\!\perp u = X$, so X^2, r_Y^2 are independent.

(b)

Step 1. $c = \frac{1}{3}/\frac{2}{3} = \frac{1}{2}$, $r_X = \frac{1}{2}(u - v)$, $Y = u + v$, $\text{Cov}(r_X, Y) = \frac{1}{2}(\mathbb{E}u^2 - \mathbb{E}v^2) = 0$.

Step 2. $(u - v)^2 (u + v)^2 = (u^2 - v^2)^2$, so

$$\mathbb{E}(r_X^2 Y^2) = \tfrac{1}{4}\mathbb{E}(u^4 - 2u^2v^2 + v^4) = \tfrac{1}{4}(\tfrac{2}{5} - \tfrac{2}{9}) = \tfrac{2}{45},$$
$$\mathbb{E}r_X^2 \, \mathbb{E}Y^2 = \tfrac{1}{4} \cdot \tfrac{2}{3} \cdot \tfrac{2}{3} = \tfrac{1}{9}, \qquad \text{Cov}(r_X^2, Y^2) = \tfrac{2}{45} - \tfrac{5}{45} = -\tfrac{1}{15}.$$

Step 3. $\mathbb{E}(u \pm v)^4 = \mathbb{E}u^4 + 6\mathbb{E}u^2\mathbb{E}v^2 + \mathbb{E}v^4 = \frac{16}{15}$, $\text{Var}\{(u \pm v)^2\} = \frac{16}{15} - \frac{4}{9} = \frac{28}{45}$, so $\text{Var}(Y^2) = \frac{28}{45}$, $\text{Var}(r_X^2) = \frac{7}{180}$ and

$$\text{Corr}(r_X^2, Y^2) = \frac{-1/15}{\sqrt{(7/180)(28/45)}} = \frac{-1/15}{14/90} = -\frac{3}{7} \approx -0.43,$$

matching the simulated -0.42. ■

Exercise 12.3 (The DirectLiNGAM recursion)

In a LiNGAM $X = BX + \epsilon$ with $A = (I - B)^{-1}$, let X_j be exogenous and $r_i = X_i - a_{ij}X_j$ for $i \neq j$. Show that

$$r_i = \sum_{k \neq j} b_{ik} r_k + \epsilon_i \qquad (i \neq j),$$

i.e. $(r_i)_{i \neq j}$ is a LiNGAM with coefficient matrix B with row and column j deleted, the same errors $(\epsilon_i)_{i \neq j}$, and the induced causal order.

Solution [Supplement].

Step 1. $A = I + BA$: $a_{ij} = \delta_{ij} + \sum_k b_{ik} a_{kj}$. Row j of B is zero, so $X_j = \epsilon_j$, $a_{jj} = 1$, and $a_{ij} = b_{ij} + \sum_{k \neq j} b_{ik} a_{kj}$ for $i \neq j$.

Step 2. With $X_j = \epsilon_j$, $X_k = r_k + a_{kj}\epsilon_j$:

$$X_i = b_{ij}\epsilon_j + \sum_{k \neq j} b_{ik}(r_k + a_{kj}\epsilon_j) + \epsilon_i = \sum_{k \neq j} b_{ik} r_k + a_{ij}\epsilon_j + \epsilon_i;$$

subtract $a_{ij}X_j = a_{ij}\epsilon_j$.

Step 3. The reduced B stays strictly lower triangular in the induced order and $(\epsilon_i)_{i \neq j}$ stay independent non-Gaussian, so the LiNGAM assumptions hold and the exogeneity characterization applies again. ■